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Additive manufacturing enhanced sidearm

Abstract

An additive manufactured enhanced sidearm comprising an ejectable barrel group assembly having two barrels, each with a chamber, in an under and over configuration thereby forming a top and bottom, and preloaded with keyed rounds having a series wherein each of said projectiles is keyed via a keyway in a chamber in said barrel, each of said projectiles in said barrel group are keyed having 0 to 4 keyways formed in the side of the projectile said projectiles having a propellant and primer means disposed behind said projectile in said chamber, a series of electrical contacts disposed within said chambers in communication with said primer means, a rail disposed at the bottom of the barrels with a locking notch means, a battery disposed with said ejectable barrel group, a grip assembly in communication with said ejectable barrel group assembly said grip assembly comprising a hand grip, a pc board disposed within said hand grip, a trigger group disposed upon said pc board, a high voltage generator circuit disposed on said pc board, at least one recoil plate assembly disposed upon said pc board, at least one microprocessor disposed upon said pc board, and a removable safety block that interferes with the batter to pc board connection.

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Background/Summary

BACKGROUND

Technical Field

(1) The embodiments herein generally relate to a side arm and more particularly to a unique sidearm created using additive manufacturing materials and techniques.

Description of the Related Art

(2) Conventional firearms for the military are manufactured years before they are ever issued to soldiers who will be using them. These firearms are made in highly specialized factories. When the military deploys to a conflict they must bring their equipment, including weapons, with them or have it shipped to them. With regard to sidearms in particular, it is common that the number of sidearms required by any unit be it a brigade or otherwise is not the number that is usually available on hand. This leads to officers including colonels and general staff being required to carry bulky M4 carbines or similar rifles in theatre when a sidearm would be sufficient for personal defense.

(3) Additive manufacturing technology has seen amazing growth and improvement over the past decades. While various components of firearms can be manufactured utilizing additive manufacturing techniques, conventional sidearm designs do not lend themselves to a high degree of additive manufacturing. Moving components such as slides and springs along with ammunition do not lend themselves to additive manufacturing processes and techniques. Similarly, designs that could lend themselves to additive manufacturing, such as designs where projectiles and propellent are pre-stacked in the barrel of the weapon have been found to be unsuitable due to their low projectile velocity and lack of lethality as compared to conventional modern firearms. One recent

failed concept was known as Metal Storm Limited (MSL). The company used the technology of electronic ignition and stacked projectiles. Unfortunately, the system was unable to obtain sufficient velocities for its projectiles to be competitive to that of modern conventional ammunition. The MSL system also suffered from mechanical and electrical unreliability issues as well as other problems. Throughout history various and sundry weapon related inventions have been created that superpose projectile load arrangement, i.e., stacking projectiles and propellants in the barrel. Some of these designs utilized multiple hammers that struck multiple primer caps in various orders. All of these weapons presented hazards and challenges to load as well as fire. Cocking the wrong hammer could end up firing the wrong charge causing damage to both the user and the weapon. The danger of a chain reaction that was caused by one round setting off the stack of rounds was a lingering fear whenever firing this type of weapon. Another issue was that of "cook offs" or a round would fire due to heat buildup in the barrel, this type of discharge of the weapon was spontaneous. All of these problems added to the list of disadvantages which prevented the idea from becoming popular even though the concept of the "superpose load", as it is often referred to in the art, in firearms existed for centuries.

(4) There have been many attempts to circumvent the disadvantages cited above. One method was to use skirts around the projectile to prevent chain reaction ignition or detonation. While the "bullet skirt" may have solved one issue, that of chain reaction ignition, it did nothing for increasing the low velocity the projectiles suffered from nor did it prevent cook offs from parasitic heat conduction. The addition of electronics for firing the rounds improved reliability somewhat but did not address other issues especially the lack of velocity. Another issue was that the recoil from firing a round would cause compression of the rounds behind the round being fired. This type of compression could cause a detonation and other failures of the system. No solution has yet been proffered to solve all the problems until now.

SUMMARY

(5) In view of the foregoing, an embodiment herein provides an additive manufacturing enhanced sidearm (AMES) comprising a firearm having an eject-able barrel assembly that incorporates a battery and two barrels in and "under and over" configuration as is commonly known in the art, said barrels containing a plurality of keyed projectiles housed in ceramic like material coated ledged chambers, propellant charges, primers and electronic means for ignition of said primers, said barrel assembly in communication with a grip assembly. The barrel assembly, in at least one embodiment are 3d printed, are formed from metal and are incased in a 3d printed housing. The chamber portion that holds the rounds is coated with a ceramic or ceramic-like material to prevent heat conduction. The frame assembly contains a locking feature to capture said barrel assembly, an unlocking and ejecting feature to hold securely and eject said barrel assembly, a recoil plate assembly to absorb the recoil transferred during firing from the barrel assembly and in communication with the barrel assembly during firing, and electronic means to electronically communicate with said barrel assembly, the frame assembly being powered by a battery disposed within the barrel assembly. In at least one embodiment of the present invention the locking and ejecting features are 3D printed compliant mechanisms, printed in place without the further addition of other parts. In other embodiments of the present invention conventional linkages and springs or a mix of conventional and compliant mechanisms are used for the barrel locking and ejecting features.

(6) The frame assembly is 3D printed along with the grips while a circuit board and trigger group is sandwiched between two 3d printed halves in communication with a recoil plugs, these assemblies are encased in a 3d printed grip housing. The circuit board and trigger group has a trigger assembly that has an articulated safety trigger, an electrical switch and a return spring along with a means for mounting said assembly to the circuit board. The trigger group being in communication with the circuit board. The recoil plate assemble is formed a of series of laminated steel plates stacked together, each of said steel plates having a plurality of apertures to accommodate the passthrough

of electrical connectors between the frame assembly and the barrel assembly. Safety features of the present invention include but are not limited to a galvanometer or continuity tester means to provide continuity testing along with a pull tab activator safety that blocks the electrical contact from the battery to the circuit board to activate the gun and make the gun functional are incorporated into the present invention. Signals from the continuity tester are sent to the microprocessor and displayed via an LED or through a holographic reflex sight. A backup hand crank generator is also integrated with a set of capacitors for emergency firing should the battery fail. In at least one embodiment a piezo electric generation system is used in case the battery fails. A “Glock” type trigger safety, as is commonly known in the art, is also incorporated into the present invention as part of the trigger group along with at least one LED that is in communication with the microprocessor assembly in order to show the readiness of the weapon. The LED may use bit angle modulation to display a variety of colors that may not be standard to a typical red/green LED.
(7) These and other aspects of the embodiments herein will be better appreciated and understood when considered in conjunction with the following description and the accompanying drawings. It should be understood, however, that the following descriptions, while indicating preferred embodiments and numerous specific details thereof, are given by way of illustration and not of limitation. Many changes and modifications may be made within the scope of the embodiments herein without departing from the spirit thereof, and the embodiments herein include all such modifications.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The embodiments herein will be better understood from the following detailed description with reference to the drawings, in which:
(2) FIG. **1** illustrates a side view of an additive manufacturing enhanced sidearm, according to an embodiment herein;
(3) FIG. **2** illustrates a cross section view of the barrel assembly of additive manufacturing enhanced sidearm of FIG. **1**, according to an embodiment herein;
(4) FIG. **3** illustrates an exploded cut away view of one of the barrels of the barrel assembly of an additive manufacturing enhanced sidearm of FIG. **1**, according to an embodiment herein;
(5) FIG. **4** illustrates a front view of the recoil plate assembly of an additive manufacturing enhanced sidearm of FIG. **1**, according to an embodiment herein;
(6) FIG. **5** illustrates a pull tab activator of an additive manufacturing enhanced sidearm of FIG. **1**, according to an embodiment herein;
(7) FIG. **6** illustrates a top view of a circuit board of an additive manufacturing enhanced sidearm of FIG. **1**, according to an embodiment herein;
DETAILED DESCRIPTION
(8) The embodiments herein and the various features and advantageous details thereof are explained more fully with reference to the non-limiting embodiments that are illustrated in the accompanying drawings and detailed in the following description. Descriptions of well-known components and processing techniques are omitted so as to not unnecessarily obscure the embodiments herein. The examples used herein are intended merely to facilitate an understanding of ways in which the embodiments herein may be practiced and to further enable those of skill in the art to practice the embodiments herein. Accordingly, the examples should not be construed as limiting the scope of the embodiments herein.
(9) An additive manufacturing enhanced sidearm **100** (hereinafter referred to as AMES) is shown in FIG. **1** in a side view. It will aid the viewer to look at FIGS. **1** and **2** simultaneously. A grip

assembly **300** and a barrel assembly **102** are shown connected to a frame assembly **200**. A holographic reflex sight **500** is attached to the frame assembly **200** as well. LED indicator(s) **600** are also shown mounted to the frame assembly **200**. Finally, the trigger **452** and trigger safety **454** are also shown connected to the frame assembly **200**.

(10) The assembly **100** and includes a barrel assembly **102** that includes two barrels **104** and **106** respectively (not depicted in FIG. **1** for clarity but shown in FIG. **2**). These barrels feature a plurality of compensating apertures **118** (shown in FIG. **2**). Each barrel **104** and **106** includes a rifled portion **110** and a chamber portion **108**. In at least one embodiment the rifled portion **110** and the chambered portion **108** are formed from one steel tubular cylinder. The inventor contemplates other embodiments wherein the barrels **104** and **106**, have separately formed chambers **108** and rifled portions **110** that are coupled to each other after forming. The inventor contemplates that the chamber portions **108** could be coupled to the rifled portions **110** via threading, welding and many other techniques known to those skilled in the art of joining metal tubular structures. Similarly, the inventor contemplates that the barrels **104** and **106**, both rifle **110** and chamber portions **108** whether formed separately or as one could be formed via metal **3***d* printing, including but not limited to powder **3***d* sintering and other methods known to those skilled in the art. A slot **242** is provided for the safety pull tab activator **240** shown in FIG. **5**

(11) An integrated holographic reflex sight **500** commonly known in the art is also provided as part of the system **100**, the sight **500** is unique since it is integrated into the microprocessor shown in FIG. **6**. The sight **500** itself is not 3d printed. The sight **500** is connected to the microprocessor **402** of the AMES **100** and actively moves the projected point of impact based on which barrel **104** and **106** is selected for firing thus ensuring that the current point of projectile impact is being correctly depicted in the sight **500**. It is further contemplated by the inventor that the projectiles **112**, propellant **114**, primers **116** and associated electrical contacts **120** are printed in place utilizing advanced additive manufacturing.

(12) Within said barrels **104** and **106** is disposed a series of recoil compensating apertures **118**, this is commonly referred to as an integral compensator **118** and each barrel **104** and **106** may have one set of these **118**. The inventor contemplates the present invention with and without this feature **118**. The inventor contemplates the integral compensator **118** can be formed in literally dozens of configurations including the one as shown in the drawings or a multitude of arrangements not shown but also known to those skilled in the art. Recoil compensators often colloquially referred to as "comps" have been known for a very long time and employed in both rifles and pistols for the purpose of reducing felt recoil to the user. A set of rails **130** is incorporated in the barrel design **102** to facilitate the sliding and locking of the barrel assembly **102** into the frame assembly **200**. A notch feature **122** in the rail **130** allows for the engagement of a locking mechanism **220** in the frame **200** to secure the barrel assembly **102** to the frame assembly **200**.

(13) Referring to FIGS. **2** and **3** or a series of keyed and stacked projectiles **112** a through **112** E for barrel **104** and **111** A through **111** E for barrel **106** respectively. The projectiles in each barrel **104** and **106** are labeled **1** through **5** in FIG. **2**. The first projectile **112** is disposed closest to the rifling **110** and the last projectile being disposed proximate to the barrel recoil plug **140**. Each projectile sits on a ledge portion **126** or flying buttress, herein after referred to as "ledge" that juts out from the inside wall **124** of the barrel. The chamber **108** is coated with a ceramic like material to prevent parasitic heat conduction. There are five ledges **126** in each barrel **104** and **106** one to support each projectile **112** respectively. Projectiles **112***b* through **112***e* will have key ways **128**, hereinafter referred to as "keys" **128** formed in them to clear and move past the ledges **126** directly in front of them. The ledge **126** prevents the projectiles **112** et al., in the stack from moving rearward when the recoil force of an initiated projectile **112** acts upon it during firing. As a round is fired the projectile **112** moves forward engaging the rifled portion **110** of the barrel **104** and **106** to impart spin to the projectile **112** and then leaves the barrel assembly **102** with great velocity. The next round **112** is then selected by the microprocessor **402** and awaits initiation by a trigger **452** pull. During firing,

the rearward force from the projectile **112** moving in the opposite direction acts against the remaining projectiles **112** in the chamber portion **108** of the barrel **104** and **106** and attempts to force them rearward. The rearward motion of the projectile **112** in the chamber **108** is arrested by the ledge **126**. This force is transferred to the barrel assembly **102** via the ledge **126** and in turn to the recoil plate **800** and the frame assembly **200**, felt recoil is reduced via the integral compensators **118** the resulting reduced recoil is then transferred to the hand of the user (not shown). FIG. **4** shows the front view of the recoil plate assembly **800**. The plate **801** houses a slot **806** for mechanically connecting the circuit board **400**. Also shown are the positive firing contacts **802**, the barrel ejector aperture **804** and the negative firing contact **808**.

(14) Propellent charges **114** through are also shown. Within these barrels **102** are the series of electrical traces **120** that extend along the inside of the barrels **104** and **106**. Each of said traces **120** being in communication within individual primer **116** and terminating at a barrel recoil plug **140** disposed at the rear end portion of the barrel **104** and **106**. Electrical traces **120** are bonded to the interior chamber portion wall **124** using an insulating nonconductive adhesive (not shown), such adhesives are commonly known in the art.

(15) A plurality of electrical traces **120** is disposed longitudinally within each barrel chamber **108** section and parallel to each other. Each trace **120** conducts electricity while being electrically insulated from the metal barrel **104** wall and other traces **120**. Each trace **120** extends from the barrel recoil plug **140** located at the rear of the barrel **104** to a specified projectile **112**. The electrical traces **120** make contact with the projectiles **112** surface. At least a portion of the surface of the projectile **112** is electrically conductive. Conductivity can be provided by a conductive coating including but not limited to any conductive material such as copper electroplating, jacketing, paint, powder coating, printing, etc., conductive materials being well known in the art the inventor contemplates many different elements as well as alloys being used for this purpose. Within the tip of the second through fifth projectiles **112** is an electrically initiated primer **116**. Note that the primer **116** can be initiated by low voltage in at least one embodiment and high voltage induced plasma in at least another embodiment. The electrical trace **120** carries a positive electrical charge to the primer **116** when the AMES **100** is fired a negative lead **132** is also in contact with the primer **116** such that a primer **116** can be initiated when the trigger **452** is pulled and the circuit is completed. The negative lead **132** is electrically insulated from positive electrical trace contacts **120**. The inventor contemplates in at least one embodiment of the present invention a single negative lead **132** extending through the stack of projectiles **112**, the projectiles **112** having a composite structure that allows the negative lead **132** to be insulated from the rest of the projectile **112**.

(16) Selection of the projectile **112** firing sequence is made by the microprocessor **402**. The microprocessor **402** selects each projectile **122** to be fired. Upon the firing of a projectile **112** the next subsequent projectile **112** is selected for firing. Typical programming of the microprocessor **402** designates the top barrel **104** for firing followed by the bottom barrel **106**. However, it is contemplated that this firing sequence can be changed by either the manufacturer or the user. Programming is facilitated via a port (not shown) located in the frame assembly **200**. Firing sequences could be but are not limited to the following top/bottom; bottom/top; all top followed by all bottom; all bottom followed by all top and any and all combinations thereof. The microprocessor **402** can detect a failed around and can automatically switch to a different barrel **104** and **106** assuming there are still projectiles **112** in said alternate barrel **102** and **104**. The integrated holographic reflex sight **500** that is slaved to the AMES system **100** detects and adjusts the appropriate point of impact depending on which barrel **104**, **106** is selected.

(17) FIG. **6** shows the trigger group **450** is comprised of a trigger **452** including a trigger safety tip **454** that is required to be actuated before the rest of the trigger **452** can be pulled. This form of trigger safety is commonly known in the art. A trigger switch **456** in communication with the trigger **452** and is activated by the pulling of the trigger **452** and associated hardware and linkage to

secure said trigger **452** and switch **456** to the circuit board assembly **400**. The electronic switch **456** sends a signal to the microprocessor **402** to fire the selected projectile **112** from the designated barrel **104**,**106**.

(18) Within the barrel assembly **102** is a battery **134** that powers the AMES **100**. When a barrel assembly **102** is inserted into the frame assembly **200** and locked in place an electrical connection is formed, the battery **134** is now in communication with the microprocessor **402** and a high voltage generation circuit **404**. By pulling the trigger **452** the user will activate an electrical switch **456** that actuates the firing circuit **404** and sends a high voltage current back to the barrel assembly **102**. The current travels along the barrel **104**, **106** via the barrel traces **120** from the recoil plate assembly **800** to the plug assembly **140** and finally to the assigned projectile **112** and its primer **116**. When actuated the primer **116** detonates and initiates burning of the propellent mixture **114** that sends the projectile **112** out of the front end of the barrel **104**, **106**. FIG. **1** shows the caliber of the round "9 MM—KALB" it should be noted that the inventor contemplates the use of many different calibers and is not limited to one specific caliber. For the present invention the changes of calibers are as easy as changing barrel assemblies. Both powder propellant and solid propellant similar to that used in caseless ammunition is contemplated by the inventor for all embodiments of the present invention. The inventor contemplates that at least in one embodiment that the solid propellant will be deposited in place using advanced **3***d* additive manufacturing. A sealant (not shown) is used between each projectile **112** to prevent chain firing and cook-offs. The sealant acts as both a physical and thermal barrier between projectiles **112** and their associated propellants **114**.

(19) Inventor contemplates the use of both standard primers **116** or electrical fired primers **116** to initiate propellant **114** burn. Electrical primers **116** commonly available for "Remington E" rifles and other known are electric primers are contemplated for use by this invention. The primers **116** are mounted in a hollow tip **113** of every subsequent projectile **112** following the first projectile **112** in the barrel **104**, **106**. The type of projectile for the first seated projectile closest to the rifling, previously referred to as "projectile **1**" can be any type of standard projectile including but not limited to; full metal jacket, exposed soft point, hollow point, ballistic tip, etc. of course each barrel **104**, **106** has its own "projectile **1**". All subsequent projectiles **112** in the chamber portion **108** of the barrel **104**, **106** must have a cavity **113** or other means for holding a primer **116** in electrical communication with the barrel assembly **102** that will initiate the projectile **112** in front of it. The last projectile **112** of each barrel **104**, **106** is initiated by a primer **116** disposed in a primer plug **136**; the primer plugs **136** is in communication with the barrel recoil plug **140**. Said primer plug **136** being disposed in or in front of the barrel recoil plug **140**. The inventor envisions many embodiments where the primer plugs **136** and its associated primer **116** are separately attached or integral to the barrel recoil plug **140**.

(20) The barrel recoil plugs **140** being in communication with a recoil plate assembly **800**, both physically and electronically, the recoil plate assembly **800** transfers the recoil from the barrel assembly **102** to the frame assembly **200** after a primer **116** has been detonated, the signals having passed from the recoil plate **800** to the connector plug **140** then to the firing contact **802**. Hereinafter referred to as the firing contacts **802**. More specifically, when the barrel assembly **102** is locked into the frame assembly **200** the firing contracts **802** are in communication with the barrel connector plug **140**. Electricity is provided to the frame assembly **200** via a rechargeable battery **134** housed in the barrel assembly **102**. Battery **134** types for use in its invention include but are not limited to lithium, lithium polymer, cadmium; lead acid, etc. A plurality of triangular shaped spring-loaded electrical contacts **138** at the bottom of the battery **134** mate up with matching contacts **238** in the frame assembly **200** and are held in communication via a locking mechanism **202** that holds the barrel assembly **102** to the frame assembly **200**.

(21) In at least one embodiment, an emergency crank generator **700** is integrated to the frame assembly **200** and can be used to power up and energize at least one capacitor **704** in communication with the hand crank **700** and the circuit board **400** of the AMES **100**. In yet another

embodiment a piezo electric generator **702** is used to generate electrical means for powering the AMES **100** and initiating the primers **116** to fire the weapon. A safety tab interrupt **240**, shown more clearly in FIG. **5**, is located in between the batteries contact **138** and the frame contact **238** of the barrel assembly **102** whereby a piece of non-conductive material prevents electrical communication between the barrel assembly **102** and the frame assembly **200** a slot **242** shown in FIG. **1** allows for tab **240** to be sandwiched between the battery **134** and the frame **200**. The safety tab **240** is removed by the user prior to firing of the AMES **100**. Inventor envisions at least one embodiment where the safety interrupt tab **240** is connected to both the user's holster and the AMES **100** such that the removal of weapon from the holster pulls the tab **240** out of the weapon thus activating it for firing.

(22) FIG. **6** shows the circuit board assembly **400** has at least one microprocessor **402**, a high voltage circuit **404**, a trigger switch assembly **450**, barrel ejecting means **406**, emergency crank generator **700** or piezo generator **702** and associated electronics **706** and capacitor **704** or capacitors, a galvanometer circuit **408** and continuity tester **410** and mechanical securing means for fastening the circuit board assembly **400** to the frame assembly **200** and the left **302** and right grips **304** of the AMES **100**. The circuit board assembly **400** and associated electronic components are the only portion of the AMES **100** along with the holographic reflex sight **500** that are not contemplated as being 3d printed. The inventor envisions that the circuit board assembly **400** will be fabricated prior to the other assemblies and stored awaiting shipment to areas around the world where and when needed by the military. The barrel ejector **406** is contemplated as being affixed to the circuit board **400**. When the lock **220** is actuated to unlock, the barrel ejector **406** ejects the barrel assembly **102**. In the embodiment depicted, the illustration shows a spring-loaded ejector **406** that protrudes through the recoil plate assembly **800** to contact and force the barrel assembly **102** out of the frame assembly **200** when actuated.

(23) FIG. **1** shows at least one lowlight LED **600** is disposed within the frame assembly **200**. The LED is connected to the circuit board **400**. The LED **600** is visible to the user by being disposed below the surface of the frame assembly **200** such that the LED **600** is visible to the user while aiming the AMES **100**. Because of the low illumination and the placement of the LED **600**, the LED **600** is only visible from a limited angle of sight thus preventing the light from being seen from other angles and potentially giving away the user's position in the low light or dark conditions. In one embodiment a red LED **600** indicates a malfunction, while a yellow LED **600** indicates low battery and a green LED **600** indicates the AMES **100** is working properly. Different colors can be created by bit angle modulation to indicate the condition of the AMES **100** thus reducing the need for multiple or special LEDs. Various modes of blinking may be used to indicate other conditions including but not limited to rounds remaining, condition of the galvanometer **408** or continuity circuit **410**. While these conditions may be simultaneously displayed on the holographic reflex **500** type sight, the LED **600** acts as a backup should the sight **500** be damaged, destroyed or otherwise rendered inoperative.

(24) The foregoing description of the specific embodiments will so fully reveal the general nature of the embodiments herein that others may, by applying current knowledge, readily modify and/or adapt for various applications such specific embodiments without departing from the generic concept, and, therefore, such adaptations and modifications should and are intended to be comprehended within the meaning and range of equivalents of the disclosed embodiments. It should be appreciated that the various combinations of the features described herein may be adjusted in size and applied either serially, in parallel, or in combinations of serial and parallel configurations. It is to be understood that the phraseology or terminology employed herein is for the purpose of description and not of limitation. Therefore, while the embodiments herein have been described in terms of preferred embodiments, those skilled in the art will recognize that the embodiments herein may be practiced with modification within the spirit and scope of the appended claims.

Claims

1. An additive manufacturing enhanced sidearm comprising: an ejectable barrel group assembly having two barrels, each with a chamber, in an under and over configuration thereby forming a top and bottom, and preloaded with keyed rounds each having a series of circumferential keyways, wherein each of said projectiles is keyed via a keyway in a chamber in said barrel, each of said projectiles in said barrel group is keyed having 0 to 4 keyways formed in the side of the projectile, said projectiles having a propellant and primer means disposed behind said projectile in said chamber, a series of electrical contacts disposed within said chambers in communication with said primer means, a rail disposed at the bottom of the barrels with a locking notch means, a battery disposed within said ejectable barrel group, a grip assembly in communication with said ejectable barrel group assembly, said grip assembly comprising: a hand grip, a circuit board disposed within said hand grip, a trigger group disposed upon said circuit board, a high voltage generator circuit disposed on said circuit board, at least one recoil plate assembly disposed upon said circuit board, at least one microprocessor disposed upon said circuit board, and a removable safety block that interferes with the battery-to-circuit board connection.
2. The additive manufactured enhanced sidearm of claim 1, wherein said projectiles further comprise a negative electric conductor that is disposed through the center of all of the projectiles and is in communication with all of the projectiles.
3. The additive manufactured enhanced sidearm of claim 1, wherein said primers are disposed within a cavity of the immediately preceding projectile forming a stack, said stack excluding a first projectile which has no need for a primer, and wherein a final projectile is initiated by a primer located on a barrel recoil primer plug disposed at the rear end of the barrel.
